

A Brief History of AI

The Dartmouth Conference — 1956

AI as a formal discipline was born at a summer workshop at Dartmouth College, USA. John McCarthy coined the term 'Artificial Intelligence'. Researchers believed human-level AI was just a decade away — an optimism that would be repeatedly humbled.

Milestone: Dartmouth 1956 — AI gets its name and its first research agenda.

The AI Winters

Twice in AI history, progress stalled dramatically and funding dried up:

First Winter (1974–1980): Early rule-based systems hit the 'combinatorial explosion' — too many rules, too little compute.

Second Winter (1987–1993): Expert systems proved too expensive to maintain; hardware failed to keep pace.

Both winters were recovery periods that forced researchers to rethink fundamentals.

Deep Blue vs. Kasparov — 1997

IBM's Deep Blue defeated world chess champion Garry Kasparov in a six-game match — the first time a computer beat a reigning world champion under tournament conditions. The event captured global attention and signalled that machines could master complex, rule-bound domains.

Milestone: Deep Blue (1997) — AI demonstrates superhuman performance in a defined intellectual domain.

The Deep Learning Explosion — Post-2010

Three catalysts ignited the modern era:

- Big Data: the internet produced vast labelled datasets (ImageNet: 14 million images).
- GPUs: graphics cards repurposed for parallel neural network training.
- AlexNet (2012): deep CNN halved image-classification error — the 'Big Bang' of modern AI.

Since 2012, AI has achieved superhuman performance in image recognition, Go (AlphaGo, 2016), protein folding (AlphaFold, 2020), and natural language (GPT series, 2018–present).

AI History Timeline at a Glance

Year	Event
1956	Dartmouth Conference — AI coined
1974–80	First AI Winter
1987–93	Second AI Winter
1997	Deep Blue defeats Kasparov
2006	Hinton reintroduces deep neural nets
2012	AlexNet wins ImageNet — deep learning era begins
2016	AlphaGo defeats Lee Sedol at Go
2017	Transformer architecture ('Attention Is All You Need')
2020	AlphaFold solves protein folding
2022–	Generative AI era (ChatGPT, Gemini, Claude)

Quick-Revision Questions

- What happened at Dartmouth in 1956?
- Name the two AI Winters and their causes.
- What three factors triggered the post-2010 Deep Learning explosion?